Prerequisite Tutorial 3 *Fractions & Rational Expressions*

ⓘ Properties of Fractions

ExA

(a) $a\left(\frac{c}{b}\right) = \frac{ac}{b}$ (b) $\frac{a}{b}\cdot\frac{c}{d} = \frac{ac}{bd}$ (c) $\frac{a}{c}+\frac{b}{c} = \frac{a+b}{c}$ (d) $\frac{a}{mb}+\frac{c}{md} = \frac{ad+cb}{mbd}$

Common ERROR: $a\left(\frac{c}{b}\right) \neq \frac{ac}{ab}$

Use the LCD! You do NOT want to do $\frac{a}{mb}+\frac{c}{md} = \frac{amd+cmb}{m^2bd}$

(e) $\frac{1}{b}\cdot\frac{1}{c} = \frac{1}{bc}$ **BUT** $\frac{1}{b}+\frac{1}{c} \neq \frac{1}{b+c}$

(f) $\frac{ab}{b} = a$ **BUT** $\frac{a+b}{b} \neq a$

(g) $\frac{a}{cb^{-1}} = \frac{ab}{c}$ **BUT** $\frac{a}{c+b^{-1}} \neq \frac{a+b}{c}$

ⓘ Reducing Fractions

Factor & reduce:

ExB

Proper reducing:

$$\frac{2x(x+1)}{x+1} = 2x$$

CORRECT

$$\frac{2x-(x+1)}{(x+1)} = \frac{2x-x-1}{(x+1)} = \frac{x-1}{x+1}$$

INCORRECT!

$$\frac{2x-\cancel{(x+1)}}{\cancel{(x+1)}} \neq 2x$$

NO!

Only common FACTORS reduce!

ExC Always **FACTOR** and then reduce (R!). Remember, **only common FACTORS cancel**.

1) $\frac{8x^2-6x}{6x^3} =$

2) $\frac{3x(x+1)^2 - x^2(x+1)}{x(x+1)^3}$

Your turn!

Use the idea of factoring out the GCF and then reducing.

1. Simplify: (a) $\frac{2a^2-3ab-9b^2}{2ab^2+3b^3}$ (b) $\frac{2(1+x^2)^2-8x^2(1+x^2)}{(1+x^2)^4}$

ⓘMultiplying/Dividing Fractions

ExD To **M!** 2 fractions, always **F!** 1st, **R!** 2nd, and *then* **M!** straight across.

1) $\frac{2}{9}\cdot\frac{21}{8}=\frac{1}{3}\cdot\frac{7}{4}$
$=\frac{7}{12}$

Ⓗ Always REDUCE (R!) before multiplying (M!)

R! the 21 w/the 9 & the 2 w/the 8. Do NOT multiply across to get 42/72 and *then* reduce.

2) $\frac{x^2-4}{x^2-4x+4}\div\frac{x+2}{2x-4}$

ⓘAdding Fractions

ExE To add 2 fractions, you MUST have the same denominator. Be sure to find the LOWEST common denominator (LCD).

1) $\frac{1}{6}+\frac{4}{15}=\frac{1}{2\cdot3}+\frac{4}{5\cdot3}$
$=\frac{1}{2\cdot3}\cdot\frac{5}{5}+\frac{4}{5\cdot3}\cdot\frac{2}{2}$
$=\frac{5}{30}+\frac{8}{30}$
$=\frac{13}{30}$

The only way to create the same denominator in both fractions is to MULTIPLY both top and bottom of each fraction by the appropriate number. (You are simply multiplying the entire fraction by 1, and thus not changing its value).

It's important to note that if both denominators already share a common factor, you only use it ONCE in the LCD. This is why you should not always merely multiply the 2 original denominators together to create your common denominator. For example, resist doing the following:

$\frac{1}{6}+\frac{4}{15}=\frac{15}{6\cdot15}+\frac{24}{15\cdot6}$
$=\frac{15+24}{90}$
$=\frac{39}{90}$
$=\frac{13}{30}$

NO!

Now you have to reduce by that shared factor of 3

2) $\frac{1}{x^2+x}-\frac{2}{x^2-1}=$

3) $\frac{x}{5x+2}+\frac{3}{5x}=\frac{x}{5x+2}\cdot\frac{5x}{5x}+\frac{3}{5x}\cdot\frac{5x+2}{5x+2}$
$=\frac{5x^2+15x+6}{5x(5x+1)}$

NO!
$\frac{3}{5x}\neq\frac{3+2}{5x+2}$

You cannot *add* the same number to the top and bottom of a fraction!

Your turn!

2. Simplify: (a) $\frac{x}{x+1}+\frac{2}{x^2+4x+3}$ (b) $\frac{3}{x^2-1}-\frac{2}{1-x}+\frac{1}{x}$ (Hint: change the $1-x$ to $-(x-1)$)

ⓘ Simplifying Fractions

ExF "Invert & Multiply"

1) $\dfrac{4}{1/x} = 4 \cdot \dfrac{x}{1}$

$= 4x$

2) $\dfrac{4/5}{x} = \dfrac{4}{5} \cdot \dfrac{1}{x}$

$= \dfrac{4}{5x}$

ExG A simplified fraction is a REDUCED fraction containing a SINGLE fraction bar.

1) $\dfrac{8+\frac{1}{x}}{16-\frac{2}{x}}$

2) $\dfrac{\frac{3}{h+1}-3}{h}$

3) $\dfrac{1+x^{-1}}{1-x^{-2}}$

Your turn!

3. Simplify: (a) $\dfrac{\frac{1}{x}+\frac{1}{y}}{1-\frac{1}{xy}}$ (b) $\dfrac{\frac{1}{3+h}-\frac{1}{3}}{h}$

Prerequisite Review problems

FYI: You will be required to show your work in the same manner as shown in this tutorial. Be sure to read the HW Guidelines *carefully*.

Be sure to read/study **ExA-D** before trying **PT3#1**!

PT3#1 Simplify. (a) $3(\frac{2}{5})$ (b) $\frac{5}{6}\cdot\frac{4}{15}$ (c) $\frac{2x^2+8x}{8x}$

(d) $\frac{8x(x+2)^2-6x^2(x+2)}{6x^3(x+2)^6}$ (e) $\frac{x^2-3x}{x^2-4}\div\frac{2x}{x+2}$

PT3#1 (a) 6/5 (b) 2/9 (c) $\frac{x+4}{4}$

(d) $\frac{x+8}{3x^2(x+2)^5}$ (e) $\frac{x-3}{2(x-2)}$

Be sure to read/study **ExE** before trying **PT3#2**!

PT3#2 Simplify and determine domain if applicable.

(a) $\frac{17}{20}-\frac{7}{8}$ (b) $\frac{2}{5x+1}+\frac{3}{5x}$ (c) $\frac{x^2-3}{x^2-x}-\frac{x+2}{2x}$

PT3#2 (a) -1/40

(b) $\frac{25x+3}{5x(5x+1)}$

(c) $\frac{x^2-x-4}{2x(x-1)}$

Leave your denominators FACTORED!!

Be sure to read/study **ExF-G** before trying **PT3#3**!

PT3#3 Simplify.

(a) $\frac{48}{8/3}$ (b) $\frac{3/20}{15}$ (c) $\frac{1}{1+\frac{3}{h+1}}$ (d) $\frac{\frac{1}{2+x}-\frac{1}{2}}{x}$

PT3#3

(a) 18 (b) 1/100 (c) $\frac{h+1}{h+4}$ (d) $\frac{-1}{2(2+x)}$

Your Turn answers

1.(a) $\frac{a-3b}{b^2}$ 2.(a) $\frac{x+2}{x-3}$ 3. (a) $\frac{y+x}{xy-1}$

(b) $\frac{2(1-3x^2)}{(1+x^2)^3}$ (b) $\frac{3x^2+5x-1}{x(x-1)(x+1)}$ (b) $\frac{-1}{3(3+h)}$